Computer Organisation

Register

Addressing Mode

- addressing modes refer to the various ways in which the operand (the data or value) for an instruction is specified. Each addressing mode defines a way of interpreting or calculating the effective address of an operand, and it plays a crucial role in how the processor accesses data during program execution.
- Opcode is typically a binary or hexadecimal value that tells the CPU what operation to execute.ex. ADD,SUB,MUL,LOAD
- The operand is the part of the instruction that specifies the data or location upon which the operation (specified by the opcode) is to be performed Ex. Immediate,Register,Memory Address,Addressing Mode
- It shows the location of a required instruction in the computer.

Types

- Immediate addressing mode:- #/I
- Register addressing mode:- REGISTER NAME
- Direct addressing mode:- [ ]
- Indirect addressing mode:-@/( )
- Indexed addressing mode:- INDEX REGISTER NAME
- Auto indexed addressing mode:- +/-

Immediate addressing mode

- This mode is used to access the constant.
- In this mode data is present in the address field of an instruction.

Register Addressing mode

- In this mode data is present in the register, the corresponding register name will be mainained in the address field of an instruction as a effective address.

Direct addressing mode

- In this mode data is present in the memory the corresponding memory address will be maintained in the address field of instruction as a effective address.

Indirect addressing mode

- This mode is used to implement the pointers
- Types

1. memory indirect addressing mode
2. register indirect addressing mode

Memory indirect addressing mode

- In this mode effective address is present in the memory the corresponding address will be maintain in the address field of an instruction as a address of effective address.

Register Indirect addressing mode

- In this mode effective address is present inregister, the corresponding register name will be maintained in the address field of an instruction ans a address of affective address.

Register

- A register in computer organization is a small, fast storage location within a computer's CPU (Central Processing Unit) that is used to store data temporarily for quick access during processing.

- FGI: Might refer to a flag that is set when a certain input condition is met (e.g., a trigger signal, external input, or condition that affects the processor).

- FGO: Might refer to a flag that is used when an output or signal condition is met, potentially indicating the result of an operation or a state change (e.g., an interrupt request, status change, or output event).
- FGI: A "Group Input" flag, indicating whether a group of input conditions or signals are active.
- FGO: A "Group Output" flag, indicating the status of output operations or signaling.
- FGI could be a flag that indicates an interrupt request or an event that triggers an interrupt.
- FGO could be a flag used to signal that the interrupt handling process has completed or a condition related to interrupt processing has been met.

Logic Gates

- A gate can be defined as a digital circuit that can allow a signal (electric current) to pass or stop.
- A truth table is a table that shows all possible combinations of inputs and outputs for a logic gate.

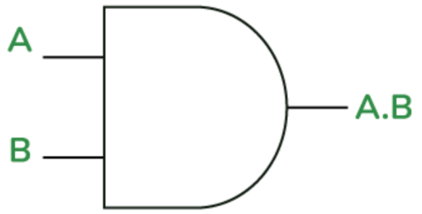
Types of Gate

- AND Gate(.)
- OR Gate(+)
- NOT Gate(‘)
- XOR Gate
- NAND Gate
- NOR Gate
- XNOR Gate
- Buffer Gate
- Universal Logic Gates

AND Gate

- The AND gate gives an output of 1 when if both the two inputs are 1, it gives 0 otherwise.
- X=(A.B)

2- Input AND Gate



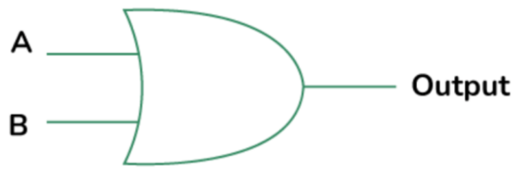
Truth Table

A (Input 1)	B (Input 2)	X = (A.B)
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate(+)

- The OR gate gives an output of 1 if either of the two inputs are 1, it gives 0 otherwise.
- X=A+B

2-Input OR Gate

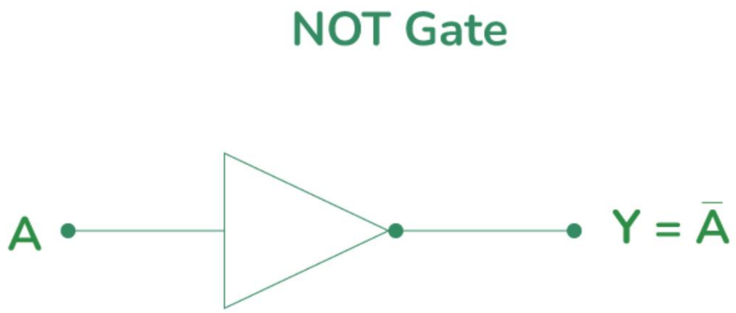


Truth Table

Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1

NOT Gate(‘)

- The NOT gate gives an output of 1 if the input is 0 and vice-versa.
- Y= A compliment

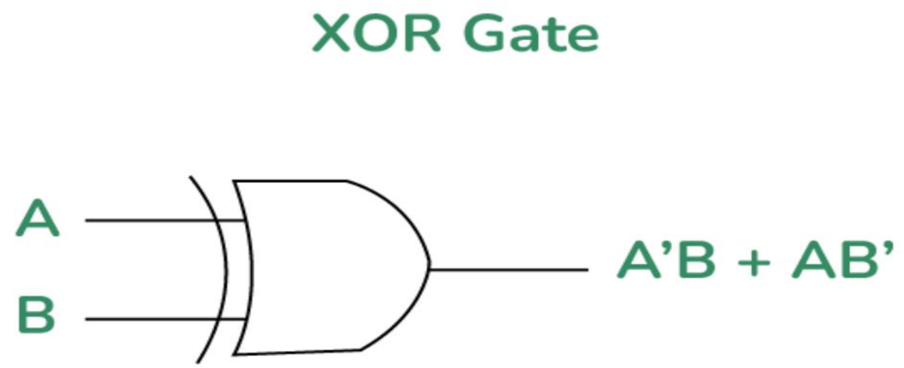


Truth Table

A (Input)	Y = $\bar{A}$ (Output)
0	1
1	0

XOR Gate

- The XOR gate gives an output of 1 if either both inputs are different, it gives 0 if they are same.
- X = A’B + AB’
- the output is 1 when there among the inputs



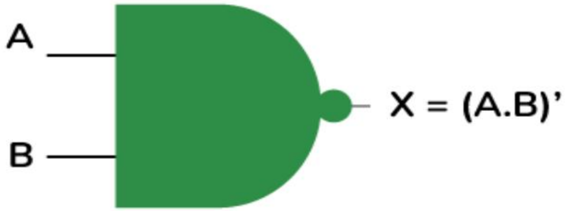
Truth Table

A (Input 1)	B (Input 2)	X = A‘B + AB’
0	0	0
0	1	1
1	0	1
1	1	0

NAND Gate

- The NAND gate (negated AND) gives an output of 0 if both inputs are 1, it gives 1 otherwise
- Y= (A.B)’

2-Input NAND Gate



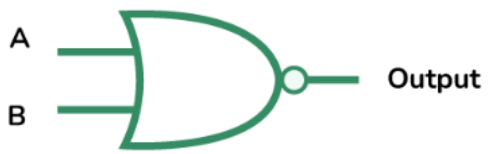
Truth Table

Input A	Input B	X = (A.B)’
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate

- The NOR gate (negated OR) gives an output of 1 only if both inputs are 0, it gives 0 otherwise.
- Y= (A+B)’

2- Input NOR Gate

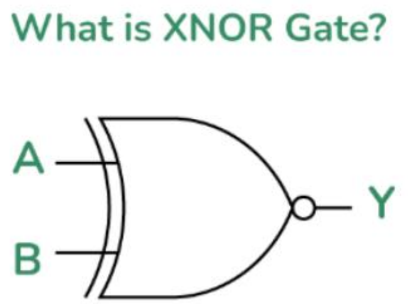


Truth Table

Input A	Input B	0 = (A + B)’
0	0	1
0	1	0
1	0	0
1	1	0

XNOR Gate

- The XNOR gate (negated XOR) gives an output of 1 if both inputs are same and 0 if they are different.
- $Y = A \odot B$
- The output is 1 when th er of 1s in the inputs



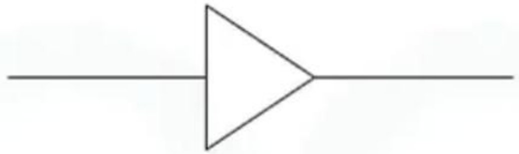
Truth Table

Input A	Input B	Output
0	0	1
0	1	0
1	0	0
1	1	1

Buffer Gate

- A Buffer Gate is a digital logic gate that amplifies a signal. It doesn’t change the logic state (it outputs exactly what is input), but its primary purpose is to ensure that the signal can drive more circuits without weakening.
- If the input is 1 , the buffer outputs 1 .
- If the input is 0 , the buffer outputs 0 .
- Y=A

Buffer

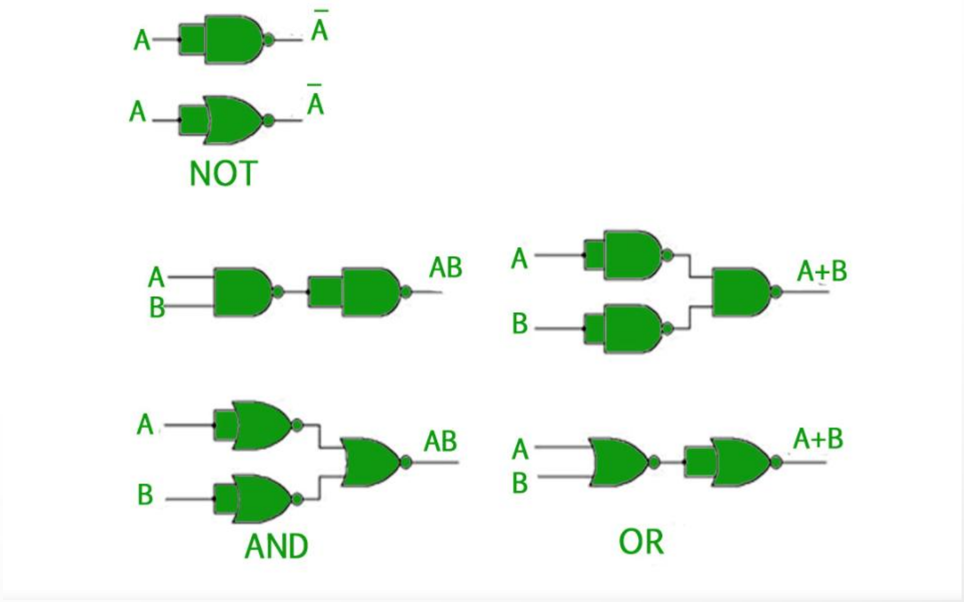


Truth Table

Input(A)	Output(Y)
0	0
1	1

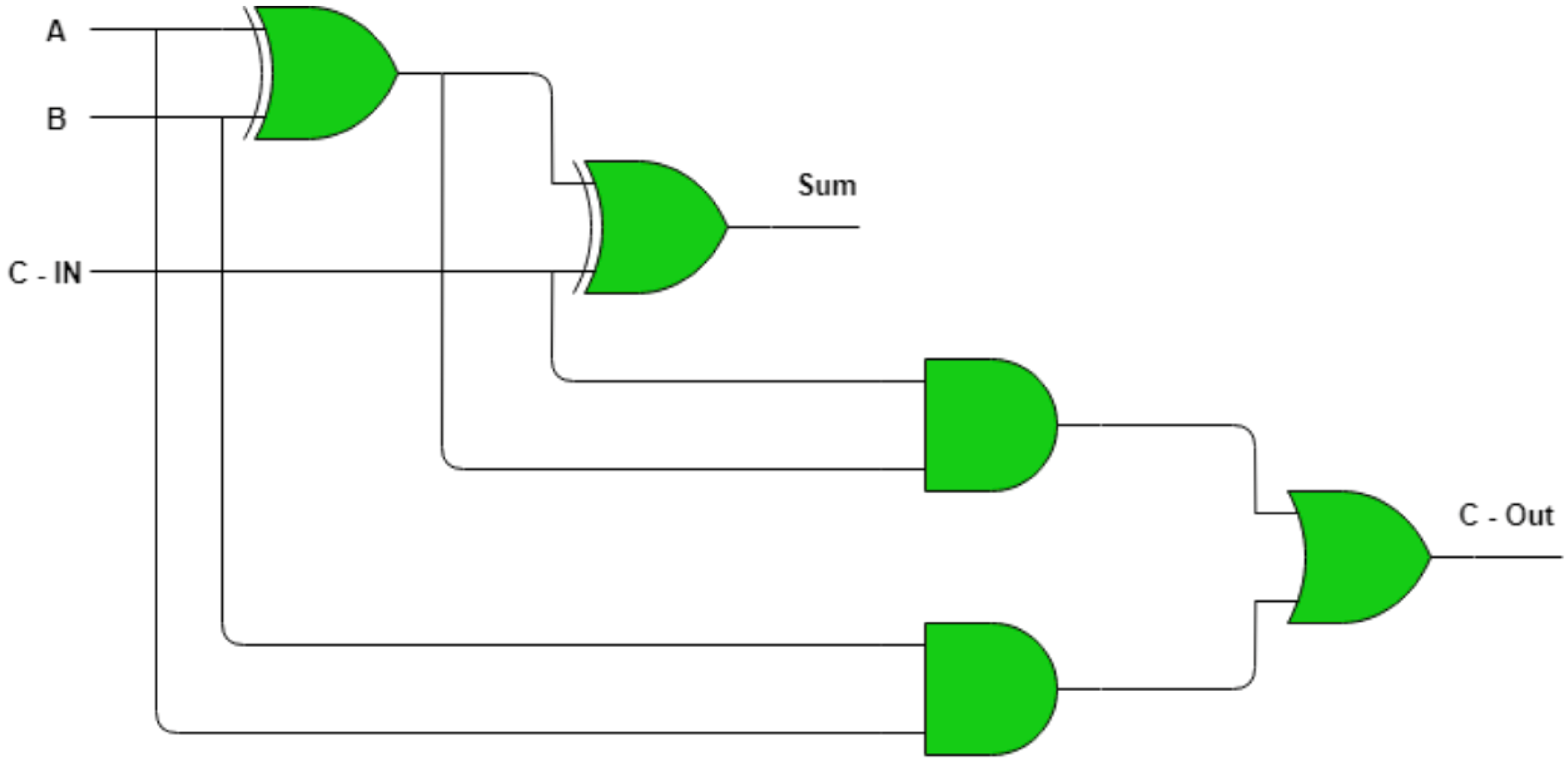
Universal Logic Gates

- Out of the eight logic gates discussed above, NAND and NOR are also known as universal gates since they can be used to implement any digital circuit without using any other gate. This means that every gate can be created by NAND or NOR gates only.



FULL ADDER

- Full Adder is the adder that adds three inputs and produces two outputs. The first two inputs are A and B and the third input is an input carry as C-IN. The output carry is designated as C-OUT and the normal output is designated as S which is SUM.



Full Adder logic circuit.

Sequencial and combination circuit

- A combinational circuit is a type of circuit where the output depends solely on the current inputs.

example Logic gates (AND, OR, NOT, NAND, NOR, XOR)

Adders (Full adder, Half adder)

A sequential circuit is different because its output depends not only on the current inputs but also on the past history of inputs.

Examples:

Flip-flops (D flip-flop, JK flip-flop, T flip-flop)

Counters (up/down counters)

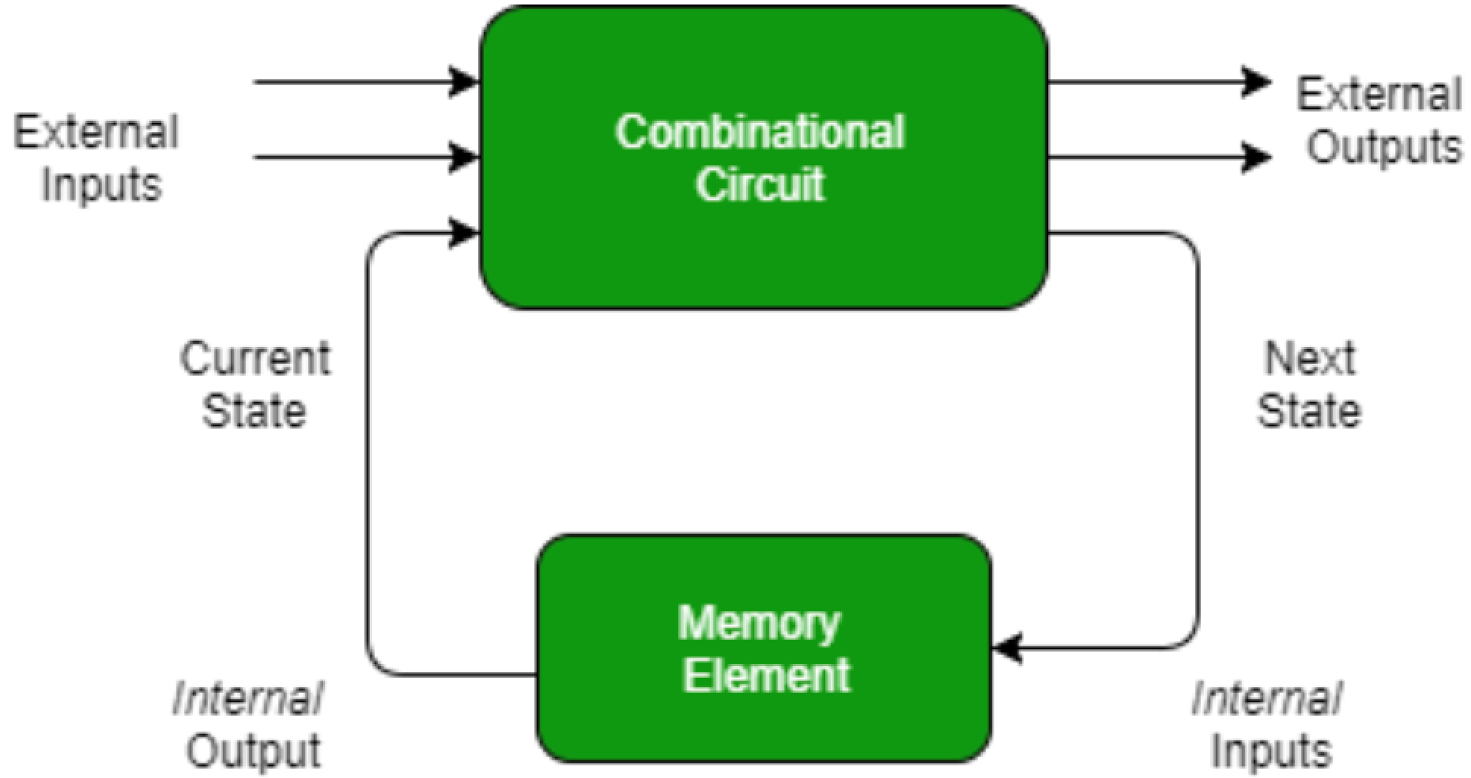


Figure: Sequential Circuit

Sequential Circuit

Flip Flop

- A flip-flop is a basic digital memory circuit that can store one bit of data.
- It’s often used in computers and electronics to remember binary states (on/off, 1/0).
- It has its two states as logic 1(High) and logic 0(low) states.

Types of Flip-Flops

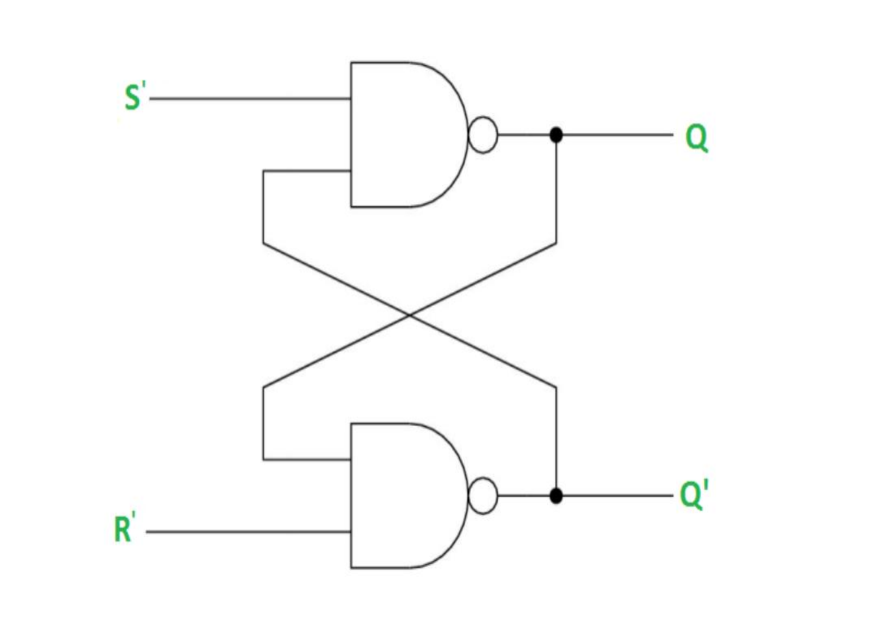
- SR Flip Flop
- JK Flip Flop
- D Flip Flop
- T Flip Flop

- The SR flip-flop (Set-Reset flip-flop) is one of the simplest types of sequential logic circuits. It's made up of two inputs: Set (S) and Reset (R), and typically two outputs: Q and Q'

SR Latch using NAND GATE

- Latches are digital circuits that store a single bit of information and hold its value until it is updated by new input signals.

S	R	Q(N+1)
0	0	INVALID
0	1	1
1	0	0
1	1	HOLD



Truth table of SR Flip flop

- clock is trigger

S	R	Q(N+1)
0	0	HOLD
0	1	0
1	0	1
1	1	INVALID

characteristic table

S	R	Q(N)	Q(N+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	DONT CARE
1	1	1	DONT CARE

S+R’Q(N)

Excitation table

Q(N)	Q(N+1)	S	R
0	0	0	DC
0	1	1	0
1	0	0	1
1	1	DC	0

- AND Gate:
- 7408: Quad 2-input AND gate
- OR Gate:
- 7432: Quad 2-input OR gate
- NOT Gate (Inverter):
- 7404: Hex inverter (NOT gate)
- NAND Gate:
- 7400: Quad 2-input NAND gate
- NOR Gate:
- 7402: Quad 2-input NOR gate

- 7482- XNOR GATE

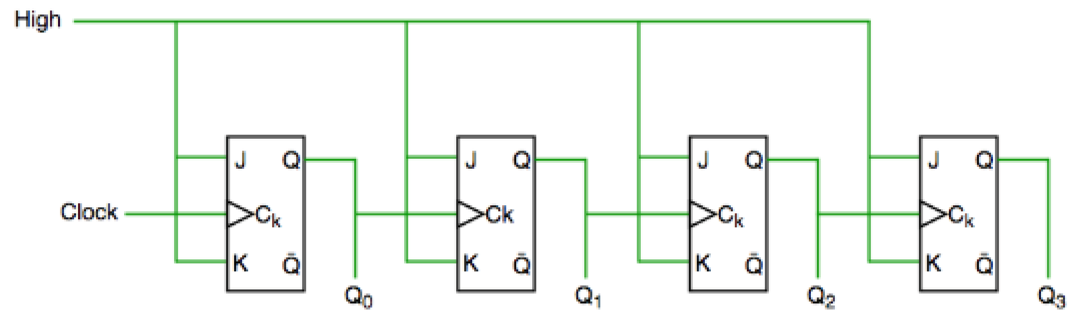
Counter

- Counter is a sequential circuit.
- A digital circuit which is used for a counting pulses is known counter.
- It is a group of flip-flops with a clock signal applied
- Counters are of two types.
- Asynchronous or Ripple Counters
- Synchronous Counters

- Classification of Counters
- Up Counters
- Down Counters

Asynchronous counter

- An asynchronous counter is a type of digital counter where the flip-flops within the counter do not receive clock pulses simultaneously. Instead, the clock pulse is applied to the first flip-flop, and each subsequent flip-flop receives its clock pulse from the output of the previous flip-flop.



(a) Asynchronous counter

Synchronous counter

- synchronous counter has one global clock which drives each flip flop so output changes in parallel
- The one advantage of synchronous counter over asynchronous counter is, it can operate on higher frequency than asynchronous counter as it does not have cumulative delay because of same clock is given to each flip flop. It is also called as parallel counter.

up and down countet

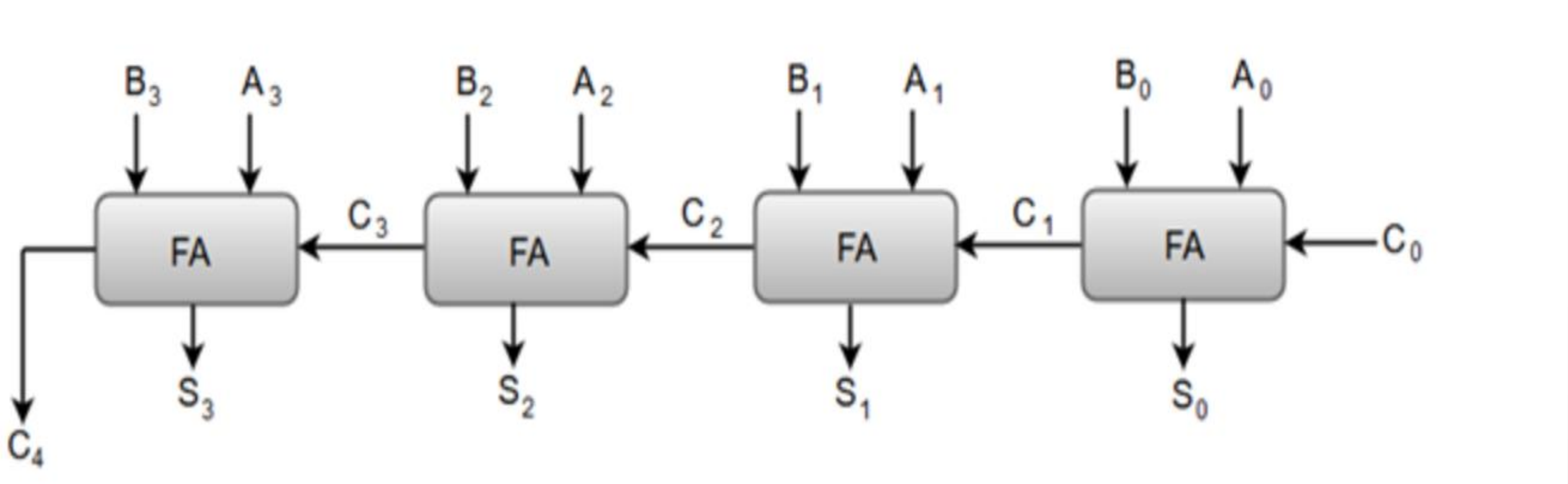
	Q(+)	Q’(-)
->(+)	down	up
-o>(-)	up	down

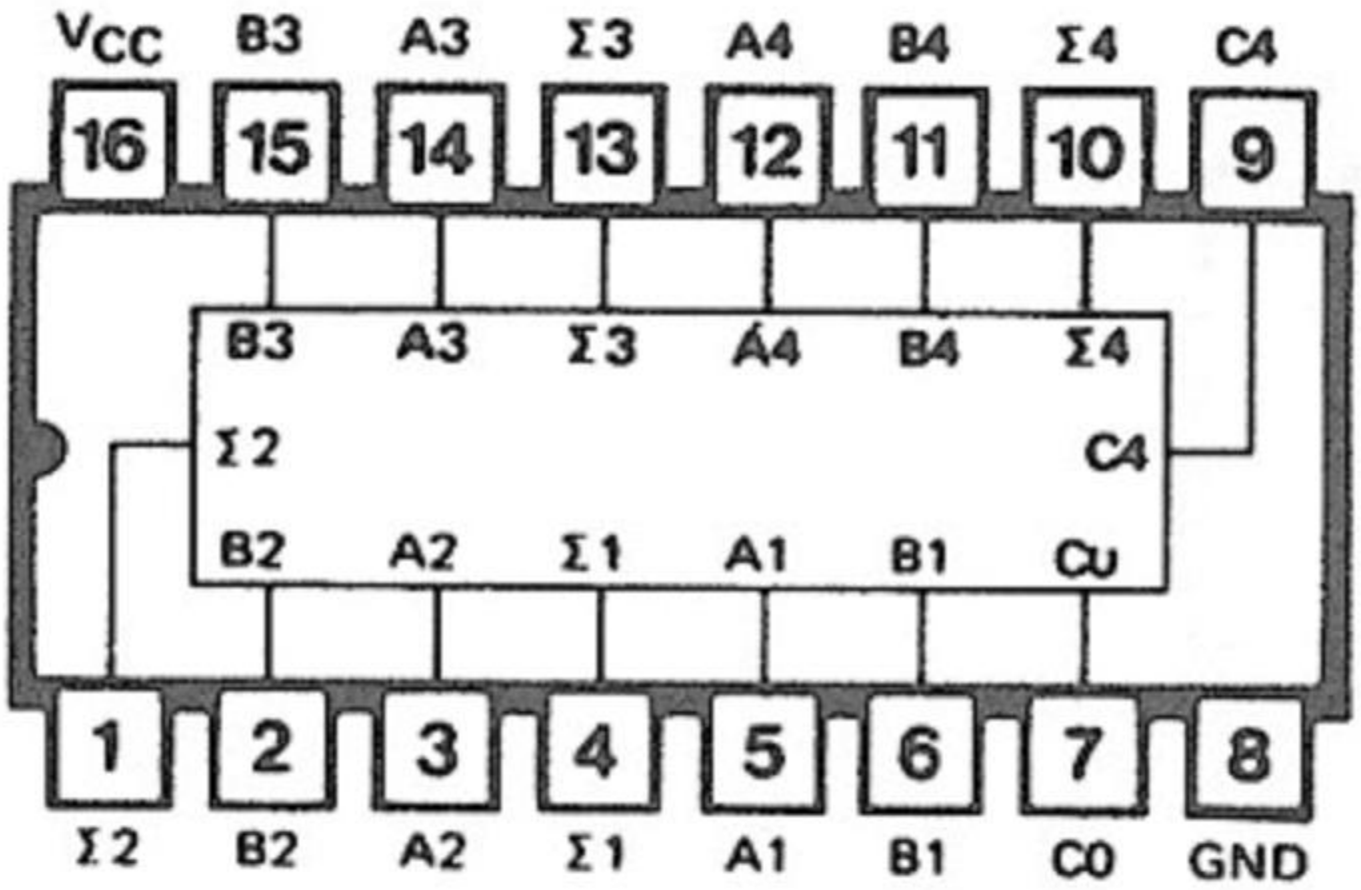
Design Synchronous counter

- 0132

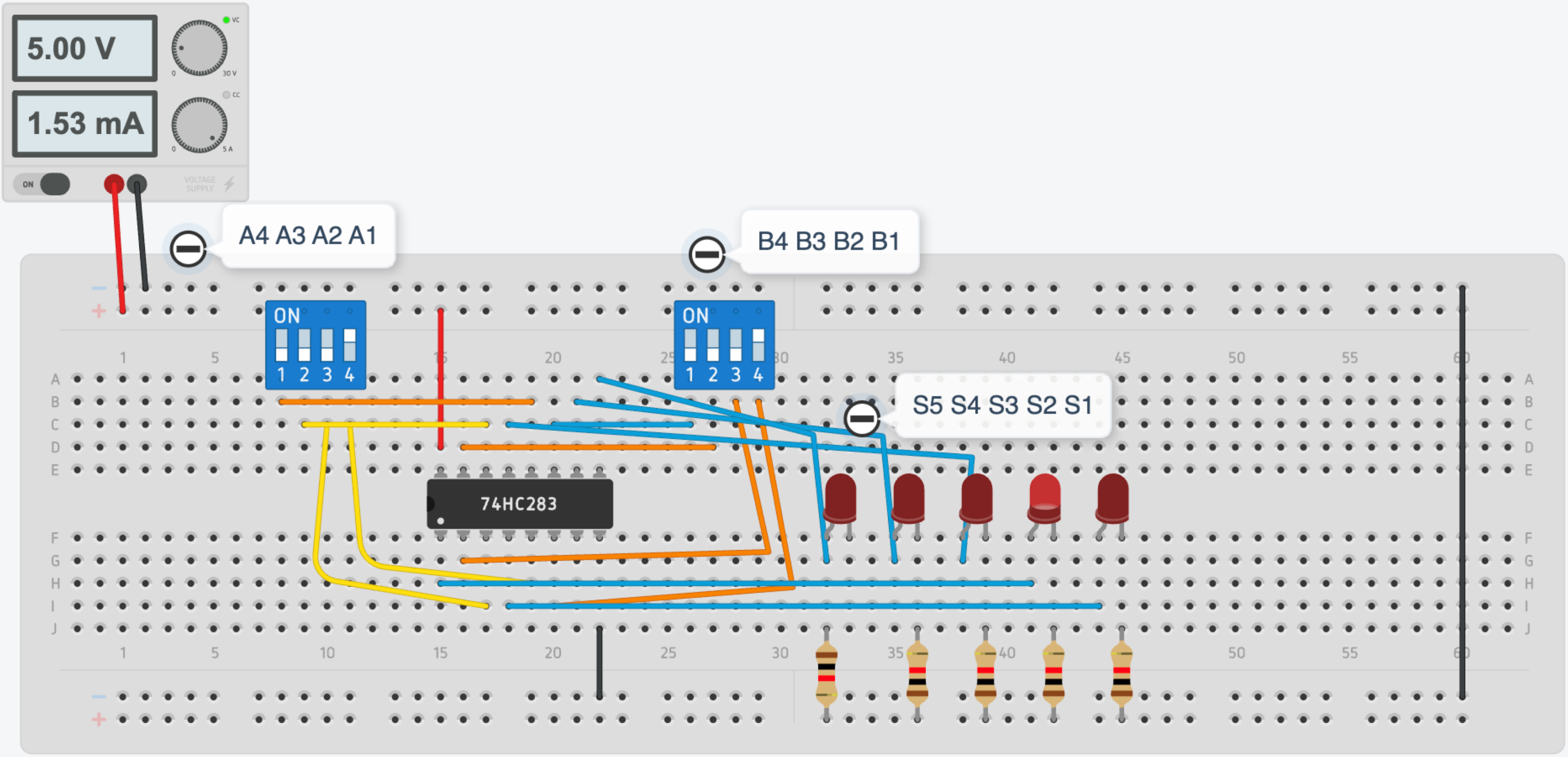
binary adder

- A Binary Adder is a digital circuit that performs the arithmetic sum of two binary numbers





V_{CC}
B3
A3
Σ3
A4
B4
Σ4
C4
16
15
14
13
12
11
10
9
B3
A3
Σ3
A4
B4
Σ4
Σ2
C4
B2
A2
Σ1
A1
B1
C0
1
2
3
4
5
6
7
8
Σ2
B2
A2
Σ1
A1
B1
C0
GND


5.00 V
1.53 mA
A4 A3 A2 A1
B4 B3 B2 B1
S5 S4 S3 S2 S1
74HC283
ON
1 2 3 4
ON
1 2 3 4